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Fault-Tolerant Model Predictive Control Trajectory Tracking for a Quadcopter with 4 Faulty Actuators

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Abstract: In this paper, an active fault tolerant control (AFTC) framework is presented, with the ability to automatically estimate and compensate for the Loss Of Effectiveness (LOE) of all actuators of a quadcopter Unmanned Air Vehicle (UAV), while achieving trajectory tracking. The aforementioned AFTC framework consists of a reconfigurable controller based on Nonlinear Model Predictive Control (NMPC), and a fault detection and diagnosis (FDD) module based on a simple and computationally cheap nonlinear algebraic observer algorithm. The presented FDD module estimates the multiplicative faults in all quadcopter actuators simultaneously in an accurate and timely manner. A nonlinear simulation validates the proposed framework and proves a precise trajectory tracking performance in the presence of 4 simultaneous actuator faults. Furthermore, simulation results show that this framework is numerically tractable and real-time applicable, as the ratio of run-time to simulation time is much less than one.

Keywords: Reconfigurable control, fault-tolerant control, model predictive control, diagnosis, fault detection, fault estimation, multiplicative faults, algebraic observer.

1. INTRODUCTION

Due to their wide-spread applications and their potential for more integration in multiple aspects in our lives, UAVs (Unmanned Air Vehicles), particularly multicopters, have been receiving a great deal of attention in both academics and industry in the recent years. Their applications ranging from delivery, to sports events filming, and drone light shows (Waibel et al. (2017)), lead to close interactions with human beings. Such close interaction increases the demand on UAVs safety and reliability. A sudden fault in an aerial vehicle during its flight could pose a risk of injury to nearby individuals. The field responsible for tackling such situation, where controllers achieve adequate performance in the presence of faults, is called Fault-Tolerant Control (FTC) (Patton (1997)).

Fault detection and diagnosis (FDD) is a crucial element in achieving fault tolerant control, since fault information provided by FDD are transmitted to the reconfigurable controller to achieve fault accommodation.

Multiple works have considered quadrotors actuator fault estimation in the literature. However, to the authors knowledge, non has showcased the estimation of 4 actuators multiplicative faults simultaneously. In (Izadi et al. (2011)), a Moving Horizon Estimation and an Unscented Kalman Filter approaches estimating 3 actuators multiplicative faults were compared. (Nguyen and Hong (2018)) utilize adaptive sliding mode Thau observer approach to estimate actuators fault in a quadrotor, however, their application is limited to only 1 actuator fault. In our approach, we are able to prove the diagnosability of, as well as estimate, all actuators multiplicative faults of a quadrotor UAV online during the flight, with a simple and computationally cheap Nonlinear Algebraic Observer (NAO).

Several methods have been investigated to design reconfigurable fault tolerant controllers, including sliding mode control (Sharifi et al. (2010); Merheb et al. (2014)), neuroadaptive FTC based on Neural Networks (Song et al. (2018)), model reference adaptive control (Chamseddine et al. (2011)), and Model Predictive Control (MPC), to name a few.

MPC in particular represents an excellent candidate for reconfigurable controllers, due to its flexible characteristics such as, constraint handling abilities for both inputs and outputs, internal dynamic model update at each time step, and incorporating nonlinear dynamics. Employing a nonlinear model as the internal model of MPC is known as Nonlinear MPC (NMPC), and allows for less model mismatch and therefore higher accuracy. Moreover, (Maciejowski (1997)) discusses implicit fault-tolerance capabilities of MPC. Further, the utilization of MPC for fault tolerant control is often referred to as Fault-Tolerant Model Predictive Control (FTMPC).

The rest of the paper is organized as follows. In Section 2, the nonlinear dynamic model of a quadcopter UAV

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is introduced, along with its faulty model. Nonlinear algebraic observer and fault diagnosability are discussed in Section 3. Section 4 discusses the use of MPC as an active fault tolerant controller. In Section 5, we validate the proposed approach in simulation for trajectory following, with a scenario including a Loss Of Effectiveness (LOE) type of fault in all actuators. Finally, conclusions are drawn in Section 6.

2. QUADCOPTER MODEL & FAULTY MODEL

A quadcopter Unmanned Air Vehicle (UAV) is a type of multicopter with 4 vertical actuators equipped with propellers.

In this section, its mathematical model is introduced briefly (Nelson et al. (1998); Samir (2007)), to describe the dynamic behavior of the system. In addition, Model Predictive control and the algebraic observer are based on the model provided, therefore, its accuracy directly affects the performance of the system.

A few assumptions are made to simplify the derivation of the system model:

- The quadcopter structure is symmetrical and rigid.
- The propellers are rigid.
- The Center of Mass (COM) and the body-fixed frame origin coincide.
- The thrust is proportional to the square of the propeller's speed.

2.1 Notations

- Vectors are represented by bolded letters.
- Upper-case letters designate matrices.
- Non-bolded, lower-case letters represent scalars.
- Superscripts B and I refer to the body and inertial frame, respectively.

For example, u is the translational speed in the body x-direction, while $\boldsymbol{u}$ is the control input vector.

2.2 Reference Frames

Two frames of reference are required to represent the quadcopter model, as shown in Fig. 1:

- Inertial or Earth-fixed frame: chosen to be a North-East-Down (NED) frame. $\mathscr{F}_I : -\{O_I, x_I, y_I, z_I\}$
- Body-fixed frame. $\mathscr{F}_B : -\{O_B, x_B, y_B, z_B\}$

The body-fixed frame $\mathscr{F}_B$ is attached to the center of mass (COM) of the quadcopter. Hence, the position of the quadcopter can be represented in the inertial frame by the position vector $\boldsymbol{\zeta} = [\, x \ y \ z \,]^T$, and its orientation is represented by the Euler angles vector $\boldsymbol{\eta} = [\, \phi \ \theta \ \psi \,]^T$ which describes the orientation of the body-fixed frame relative to the inertial frame.

2.3 Kinematics

In this subsection , the kinematics of the quadcopter system are introduced. They include two matrices. Firstly, A rotation matrix R_{IB}, that can transform the translational velocity vector in the body frame $\boldsymbol{V} = [\, u \ v \ w \,]^T$ to the time rate of change of the position vector $\dot{\boldsymbol{\zeta}}$. Secondly, a transformation matrix that transforms the body angular velocities vector $\boldsymbol{\omega} = [\, p \ q \ r \,]^T$ to time rate of change of the Euler angles $\dot{\boldsymbol{\eta}}$.

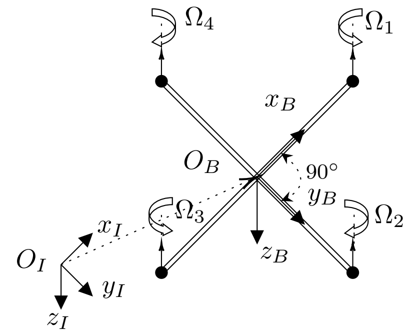

Fig. 1. Reference frames

Rotation Matrix A rotation from the inertial configuration to the final body configuration can be divided into 3 steps of yaw, pitch, and roll rotations (ψ, θ, ϕ) respectively, resulting in the rotation matrix R_{IB} from body-fxed frame to inertial frame.

$$R_{IB} = \begin{bmatrix} C_\theta C_\psi & S_\phi S_\theta C_\psi - C_\phi S_\psi & C_\phi S_\theta C_\psi + S_\phi S_\psi \\ C_\theta S_\psi & S_\phi S_\theta S_\psi + C_\phi C_\psi & C_\phi S_\theta S_\psi - S_\phi C_\psi \\ -S_\theta & S_\phi C_\theta & C_\phi C_\theta \end{bmatrix} \quad (1)$$

Where $S_\theta = sin(\theta), C_\theta = cos(\theta)$, and similarly for ϕ, ψ.

Transformation Matrix

$$\dot{\boldsymbol{\eta}} = \begin{bmatrix} \dot\phi \\ \dot\theta \\ \dot\psi \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & S_\phi T_\theta & C_\phi T_\theta \\ 0 & C_\phi & -S_\phi \\ 0 & S_\phi / C_\theta & C_\phi / C_\theta \end{bmatrix}}_{T_{IB}} \begin{bmatrix} p \\ q \\ r \end{bmatrix} \quad (2)$$

Where T_{IB} is the transformation matrix, and $T_\theta = tan(\theta)$.

2.4 External Forces and Torques

External forces acting on the quadcopter in the body frame are as follows:

$$\begin{aligned} \boldsymbol{F}_g^B &= R_{IB} \boldsymbol{F}_g^I = R_{IB} [\, 0 \ 0 \ mg \,]^T \\ &= [\, -mgS_\theta \ mgS_\phi C_\theta \ mgC_\phi C_\theta \,]^T \end{aligned} \quad (3)$$

Where $\boldsymbol{F}_g^B$ is the graviational force in the body-frame, and g is the gravitational acceleration.

$$\boldsymbol{F}_d^B = [\, -k_x u \ -k_y v \ -k_z w \,]^T \quad (4)$$

$\boldsymbol{F}_d^B$ is the drag forces vector affecting the drone in body-frame, and k_x, k_y, k_z are positive scalar translational drag coefficients.

$$\boldsymbol{F}_{th}^B = \left[\, 0 \ 0 \ -k_T \sum_{i=1}^{4} \Omega_i^2 \,\right]^T \quad (5)$$

$\boldsymbol{F}_{th}^B$ is the thrust forces vector resulting from the actuators, k_T is the thrust constant, and Ω_i is the rotational speed of the i^{th} actuator in rotations per minute (rpm). The direction of rotation of each Ω_i is shown in Fig. 1.

$$\boldsymbol{F}^B = \boldsymbol{F}_d^B + \boldsymbol{F}_g^B + \boldsymbol{F}_{th}^B \quad (6)$$

$\boldsymbol{F}^B$ is the total external force acting on the body-frame with F_x^B, F_y^B and F_z^B being its components in x, y, z body axes.

External torques acting on the system:

$$\boldsymbol{\tau}^B = \boldsymbol{\tau}_d^B + \boldsymbol{\tau}_{th}^B \tag{7}$$

$$\boldsymbol{\tau}_{th}^B = \begin{bmatrix} \tau_{th,x}^B \\ \tau_{th,y}^B \\ \tau_{th,z}^B \end{bmatrix} = \begin{bmatrix} k_T l(-\Omega_2^2 + \Omega_4^2) \\ k_T l(\Omega_1^2 - \Omega_3^2) \\ b(-\Omega_1^2 + \Omega_2^2 - \Omega_3^2 + \Omega_4^2) \end{bmatrix} \tag{8}$$

Where $\boldsymbol{\tau}_{th}^B$ is the torque resulting from the actuators thrust forces, b is the drag constant, and l is arm length.

$$\boldsymbol{\tau}_d^B = \begin{bmatrix} -k_p p & -k_q q & -k_r r \end{bmatrix}^T \tag{9}$$

$\boldsymbol{\tau}_d^B$ is the drag torque resulting from the drone's rotational motion. k_p, k_q, k_r are positive rotational drag coefficients.

2.5 Nonlinear Model

Using Newton-Euler approach, along with the external forces and torques acting on the system, a nonlinear model can be deducted as follows:

$$\begin{cases} \dot{\boldsymbol{\zeta}} = R_{IB}\boldsymbol{V} & \text{(10a)} \\ \dot{\boldsymbol{V}} = \dfrac{1}{m}\boldsymbol{F}^B - \boldsymbol{\omega} \times \boldsymbol{V} & \text{(10b)} \\ \dot{\boldsymbol{\eta}} = T_{IB}\boldsymbol{\omega} & \text{(10c)} \\ I\dot{\boldsymbol{\omega}} = -\boldsymbol{\omega} \times (I\boldsymbol{\omega}) + \boldsymbol{\tau}^B & \text{(10d)} \end{cases}$$

Where $\boldsymbol{x} = \begin{bmatrix} \boldsymbol{\zeta}^T & \boldsymbol{V}^T & \boldsymbol{\eta}^T & \boldsymbol{\omega}^T \end{bmatrix}^T \in \mathbb{R}^{n_x}$ is the state vector, with n_x the number of states. $I = diag(I_{xx}, I_{yy}, I_{zz})$ is the inertia matrix which is constant as it is referred to the body-fixed frame $\mathscr{F}_B$. In addition, it is assumed that all the states are measurable, rendering the output vector equal to the state vector $\boldsymbol{y} = \boldsymbol{x}$.

2.6 Faulty Model

To represent the model with actuator multiplicative faults, equations (10b), (10c) can be rewritten as follows:

$$\begin{cases} \dot{u} = F_x^B/m + rv - qw & \text{(11a)} \\ \dot{v} = F_y^B/m + pw - ru & \text{(11b)} \\ \dot{w} = F_{th,z}^B/m + qu - pv + gC_\phi C_\theta & \text{(11c)} \\ \dot{p}I_{xx} = \tau_{th,x}^B + (I_{yy} - I_{zz})qr & \text{(11d)} \\ \dot{q}I_{yy} = \tau_{th,y}^B + (I_{zz} - I_{xx})pr & \text{(11e)} \\ \dot{r}I_{zz} = \tau_{th,z}^B + (I_{xx} - I_{yy})pq - k_r r & \text{(11f)} \end{cases}$$

k_x, k_y, k_z, k_p, k_q terms are neglected due to their insignificance.

Equations (11c)-(11f) include control actions $\boldsymbol{u}$, and they can be rewritten in a matrix form as follows:

$$\begin{bmatrix} \dot{w}m \\ \dot{p}I_{xx} \\ \dot{q}I_{yy} \\ \dot{r}I_{zz} \end{bmatrix} = ML\boldsymbol{u} + \begin{bmatrix} m(qu - pv) + mgC_\phi C_\theta \\ (I_{yy} - I_{zz})qr \\ (I_{zz} - I_{xx})pr \\ (I_{xx} - I_{yy})pq - k_r r \end{bmatrix} \tag{12}$$

$$\begin{bmatrix} F_{th,z}^B \\ \tau_{th,x}^B \\ \tau_{th,y}^B \\ \tau_{th,z}^B \end{bmatrix} = \underbrace{\begin{bmatrix} -k_T & -k_T & -k_T & -k_T \\ 0 & -k_T l & 0 & k_T l \\ k_T l & 0 & -k_T l & 0 \\ -b & b & -b & b \end{bmatrix}}_{M} L \underbrace{\begin{bmatrix} \Omega_1^2 \\ \Omega_2^2 \\ \Omega_3^2 \\ \Omega_4^2 \end{bmatrix}}_{\boldsymbol{u}} \tag{13}$$

Where M is the allocation matrix, mapping the squared rotor speeds to the thrust force and torques in body frame, and $\boldsymbol{u} \in \mathbb{R}^{n_u}$ is the control action vector in $(rpm)^2$, with n_u being the number of control inputs.

Matrix $L = diag(l_1, l_2, l_3, l_4)$ is introduced as a diagonal matrix to represents the actuators effectiveness (Fault Parameters). The effectiveness of the ith actuator is therefore $0 \le l_i \le 1$, $(i = 1, \ldots, 4)$, where $l_i = 0$ depicts a complete actuator failure, $l_i = 1$ a healthy actuator, and $0 < l_i < 1$ a partial actuator failure (Loss of Effectiveness). It is worth noting that "faults parameters" and "actuators effectiveness" are used interchangeably in this paper.

From equations (10), (11), the nonlinear model can be represented as follows:

$$\dot{\boldsymbol{x}} = f(x, u, L) \tag{14}$$

3. FAULT DETECTION AND DIAGNOSIS

The goal of the FDD module is to detect, estimate, and isolate the faults in the system in a timely manner. Timely delivered fault information to the controller is necessary for recovery in faulty situations, which necessitates a sufficiently fast online estimator.

3.1 Algebraic Observer

The algebraic Observer being based on algebraic equations with direct substitution of outputs, inputs, and potentially their derivatives, proves to be an excellent candidate for FDD in terms of the speed of fault estimation.

It is possible to rewrite $L\boldsymbol{u}$ in equation (12) as $diag(\boldsymbol{u})\boldsymbol{\Gamma}$ (Rotondo et al. (2016)):

$$L\boldsymbol{u} = diag(\boldsymbol{u})\Gamma = \begin{bmatrix} l_1\Omega_1^2 & l_2\Omega_2^2 & l_3\Omega_3^2 & l_4\Omega_4^2 \end{bmatrix}^T \tag{15}$$

Which means that the diagonal matrix L representing the faults, can be exchanged by a vector $\boldsymbol{\Gamma}$, and the control actions vector $\boldsymbol{u}$ is transformed into a diagonal matrix $diag(\boldsymbol{u})$.

With this step, it is now possible to rewrite equation (12) to separate actuators effectiveness:

$$\begin{aligned} \hat{\boldsymbol{\Gamma}} = \begin{bmatrix} \hat{l_1} \\ \hat{l_2} \\ \hat{l_3} \\ \hat{l_4} \end{bmatrix} &= diag(\boldsymbol{u})^{-1} M^{-1} \\ &\left(\begin{bmatrix} \dot{w}m \\ \dot{p}I_{xx} \\ \dot{q}I_{yy} \\ \dot{r}I_{zz} \end{bmatrix} - \begin{bmatrix} m(qu - pv) + mgC_\phi C_\theta \\ (I_{yy} - I_{zz})qr \\ (I_{zz} - I_{xx})pr \\ (I_{xx} - I_{yy})pq - k_r r \end{bmatrix} \right) \end{aligned} \tag{16}$$

$\hat{l}_1$ to $\hat{l}_4$ in equation (16) each satisfy a differential algebraic polynomial in the form:

$$P(\hat{l}_i, \boldsymbol{y}, \dot{\boldsymbol{y}}, \ldots, \boldsymbol{u}, \dot{\boldsymbol{u}}, \ldots) = 0, \quad (i = 1, \ldots, 4) \tag{17}$$

This proves them algebraically observable, and hence diagnosable (Ibrir (2003)). Additionally, it is evident from equation (16) that in case of zero control input, actuators effectiveness cannot be determined. However, control inputs are far from zero once the propellers start rotating.

Using equation (16), actuators effectiveness can be calculated online, to be used in updating MPC internal model to maintain a stable flight and, furthermore, achieve trajectory tracking under all actuators faults.

As evident from the algebraic observer equation (16), only 8 outputs are needed to calculate the actuators effectiveness, namely, y_4 to y_8, and y_{10} to y_{12}. For 4 of which, time derivatives are required: y_6, and y_{10} to y_{12}. For noisy outputs, time derivatives are determined by the sliding mode robust differentiation technique with a smoothing element to reduce the effect of noise on the signal, see (Levant (1998)). It should be noted that the robustness and accuracy of the differentiator system utilized is essential for the success of the algebraic observer (Ibrir (2003)).

4. FAULT-TOLERANT MODEL PREDICTIVE CONTROL

MPC solves online, at each time step, a finite horizon optimal control problem. The solution returns a sequence of control actions over the future horizon. Only the first control action is selected from the sequence to be implemented on the system.

One of the advantages of MPC is its ability to update its internal system model at each sampling time. This advantage is exploited in achieving an active fault-tolerant control system. Particularly, by updating the fault parameters estimated online by the FDD module into the internal model of MPC, so that MPC calculates a stabilizing control action, that takes into consideration and compensates for faults.

It can be understood that when an actuator loses effectiveness, some of the other actuators might need to exert more effort to compensate for such loss and achieve the same desired trajectory. Their increased effort might surpass the actuators' physical limit. Luckily, another MPC advantage reveals itself to handle such situation, which is its ability to explicitly account for constraints on control inputs and system states.

Following is the Optimal Control Problem (OCP) for FTMPC:

$$\min_{\boldsymbol{U},\boldsymbol{X}} J\left(\boldsymbol{x}_k, \boldsymbol{U}, \boldsymbol{X}^{ref}\right) = \sum_{i=0}^{N_p-1} \left(\|\tilde{\boldsymbol{x}}_{k+i|k}\|_Q^2 + \|\boldsymbol{u}_{k+i|k}\|_R^2 + \|\Delta\boldsymbol{u}_{k+i|k}\|_{R_\Delta}^2\right) + \|\tilde{\boldsymbol{x}}_{k+N_p|k}\|_{Q_N}^2 \tag{18}$$

Subject to:

$$\boldsymbol{x}_{k+i+1|k} = f_d\left(\boldsymbol{x}_{k+i|k}, \boldsymbol{u}_{k+i|k}, \hat{L}(k)\right) \tag{19}$$

$$\boldsymbol{x}_{min} \leq \boldsymbol{x}_{k+i|k} \leq \boldsymbol{x}_{max} \tag{20}$$

$$\boldsymbol{u}_{min} \leq \boldsymbol{u}_{k+i|k} \leq \boldsymbol{u}_{max} \tag{21}$$

$$\boldsymbol{x}_{k|k} = \boldsymbol{x}_k \tag{22}$$

$$\text{for } i = 0, 1, \ldots, N_p - 1$$

Where f_d is determined from equation (14) by means of numerical integration. $\hat{L} = diag(\hat{\boldsymbol{\Gamma}})$ is the actuators effectiveness estimated by the algebraic observer, and N_p is the prediction horizon. $\boldsymbol{x}_{k+i|k}$ is the prediction of $\boldsymbol{x}(k+i)$ computed based on the measured value $\boldsymbol{x}_k$ at instant k, and similarly for $\boldsymbol{u}_{k+i|k}$. $\boldsymbol{x}_{k|k}$ is the initial value of the optimization problem at time instant k.

$$\tilde{\boldsymbol{x}}_{k+i|k} = \boldsymbol{x}_{k+i|k} - \boldsymbol{x}_{k+i}^{ref}$$

$$\Delta\boldsymbol{u}_{k+i|k} = \boldsymbol{u}_{k+i|k} - \boldsymbol{u}_{k+i-1|k}$$

At each instant k at the start of the receding horizon iteration, $\boldsymbol{x}_k$ and $\boldsymbol{u}_{k-1}$ are known. Where $\boldsymbol{u}_{k-1}$ is the control action from the previous iteration. $Q, Q_N \in \mathbb{R}^{n_x \times n_x}$, R and $R_\Delta \in \mathbb{R}^{n_u \times n_u}$ are diagonal positive definite weighting matrices.

$\boldsymbol{U} = \{\boldsymbol{u}_{k+i|k} | i = 0, 1, \ldots, N_p - 1\}$ is the control input sequence (input trajectory). $\boldsymbol{X} = \{\boldsymbol{x}_{k+i|k} | i = 0, 1, \ldots, N_p\}$ is the predicted state trajectory. $\boldsymbol{X}^{ref} = \{\boldsymbol{x}_{k+i}^{ref} | i = 0, 1, \ldots, N_p\}$ is the reference state trajectory.

The last term in equation (18) represents the so-called terminal cost, which plays a pivotal role in the stability of MPC. It was shown in (Mayne et al. (2000)) that utilizing MPC with a terminal cost is sufficient to achieve stability for nonlinear models, without the explicit need for a terminal constraint, provided that a long enough prediction horizon is used.

Trajectory tracking, in this work, is achieved by providing the future N_p way-points of the desired trajectory at each time step to MPC, as shown in Fig. 2. This allows MPC to use its model-based prediction capabilities to find suitable control actions to achieve the desired trajectory.

5. SIMULATION AND RESULTS

In this section, the results of the AFTC framework is demonstrated with a simulation of the system nonlinear dynamics, using MATLAB Simulink, and CasADi framework (Andersson et al. (2019)). Presented in Table 1 are Quadcopter parameters from (Alexis et al. (2012)), along with other parameters native to this paper.

Fig. 2 illustrates the FTC unit based on NMPC, and FDD module based on Algebraic Observer.

Table 1. Parameters

Parameters	values
$I_{xx} = I_{yy}$	$0.0196\ kg.m^2$
I_{zz}	$0.0264\ kg.m^2$
m	$1.1\ kg$
k_T	$9.29e-5\ N.s^2$
b	$1.1e-6\ N.m.s^2$
l	$0.21\ m$
k_r	$9\ N.m.s/rad$
g	$9.81\ m/s^2$
N_p	30

An abrupt LOE type of fault is injected into the nonlinear system, reducing the effectiveness of actuators 1, 2, 3, and 4 instantaneously to 0.9, 0.85, 0.8, and 0.75, respectively, at time $t = 10\ sec$. Fault parameters (Actuators effectiveness) are successfully estimated by the nonlinear

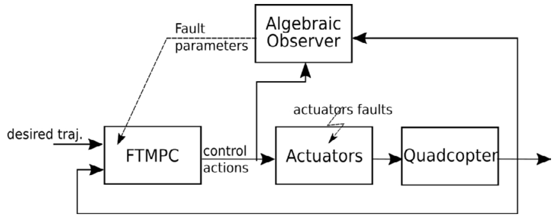

Fig. 2. System components

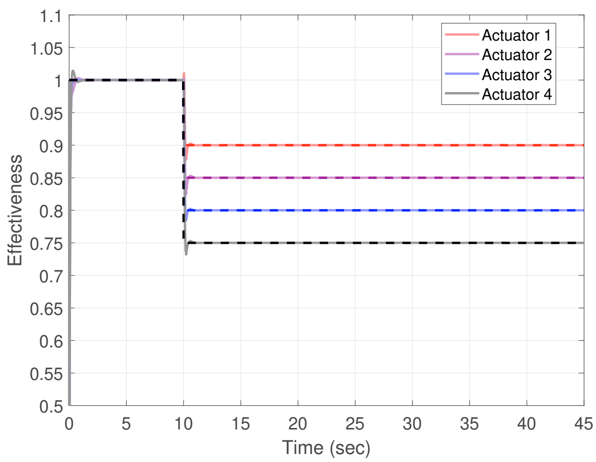

Fig. 3. estimated actuators effectiveness in transparent solid lines, actual injected effectiveness in dashed lines.

algebraic observer. The estimated faults are then injected into the FTMPC algorithm to update its internal model and mitigate fault effect on the system performance. Fig. 3 depicts an excellent estimation of the actuators effectiveness, providing an accurate and quick fault estimation.

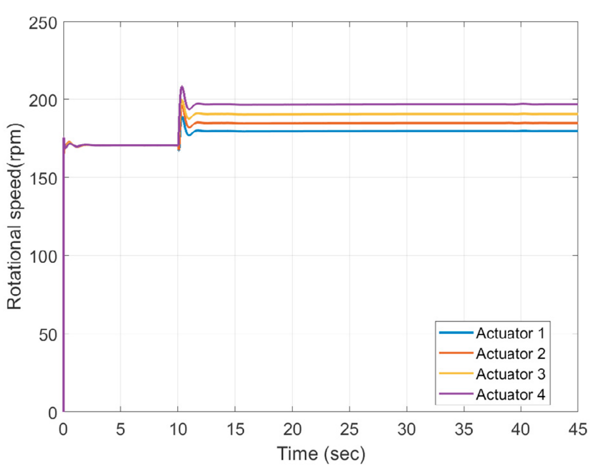

Fig. 4. Actuators rotational speed (the square root of the control action)

With the control actions being $\boldsymbol{u} = \left[\Omega_1^2 \ldots \Omega_4^2\right]^T$, Fig. 4 shows Ω_i, for convenience. Abrupt change in actuators effectiveness at the time of fault changes the MPC internal model abruptly, consequently, leading to an expected rapid change in control actions aiming to mitigate the sudden effect of LOE, as seen in the same figure. Similar effect can be seen in, for example, (Izadi et al. (2011); Jiang and Yu (2012)). An advantage of MPC is pronounced in this result, which is its ability to respect control inputs constraints even in the presence of significant faults.

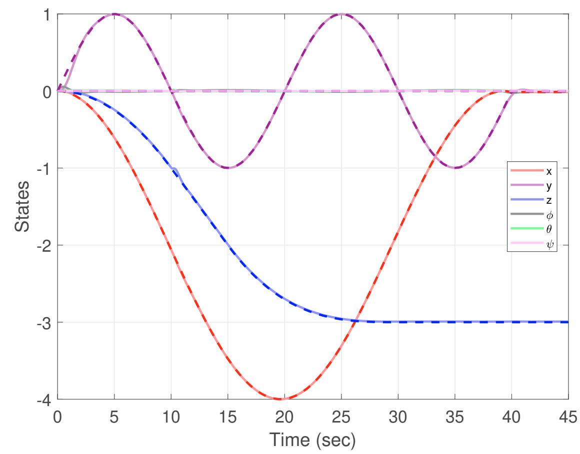

Fig. 5. States (dashed lines: desired states, transparent solid lines: actual states)

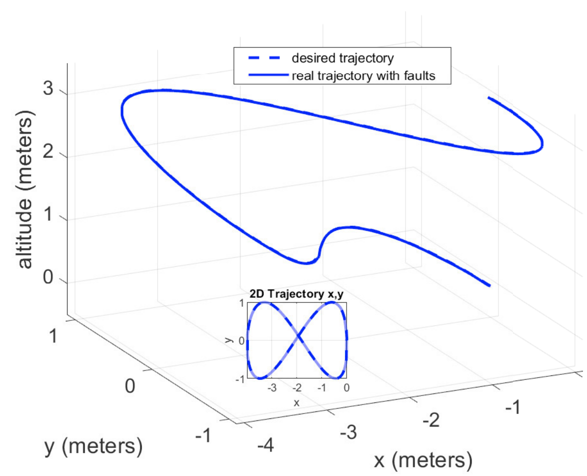

Fig. 6. 3D and 2D Trajectories

In Fig. 5, the presented framework could maintain a trajectory tracking performance of the system even in the presence of multiple faults, with minor deviations starting at the fault time $t = 10$ *sec*. The small deviations in trajectory tracking occurring at the fault time are due to fault information being incorporated in the MPC internal model 1 MPC time step later. Therefore, a larger MPC sample time would result in a delayed incorporation of fault information in the MPC internal model, resulting in higher deviation in trajectory tracking and a potentially deteriorated system performance.

A satisfying trajectory tracking performance is depicted in Fig. 6, which illustrates a 'lying eight' trajectory in 2D and 3D.

The combination of NMPC, and the nonlinear algebraic observer form an AFTC framework that is numerically tractable, and therefore, real-time applicable, with a run-time to simulation time ratio of less than one, precisely, 0.27. The overall algorithm is run with a step size of 1 ms

and MPC sampling time of 50 ms, on a desktop PC with a 3.2 GHz Intel Core (TM) i7-8700 processor and a 16GB memory (RAM).

6. CONCLUSION

In this paper, an AFTC framework is introduced. A proposed nonlinear algebraic observer successfully estimates all multiplicative faults of a quadcopter UAV, and an NMPC serves as a reconfigurable fault tolerant controller. A nonlinear simulation including the full framework is performed, validating the proposed approach, and depicting an excellent trajectory tracking performance of a 3D 'lying eight' figure, in the presence of 4 faulty actuators. Lastly, simulation results prove the proposed framework efficient, tractable, and real-time applicable, with run-time to simulation time ratio of 0.27, when run on a desktop PC with the specifications mentioned in Section 5.

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